

Nansen: Empowering crypto investors with blockchain data intelligence driven by Google Cloud

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ABOUT NANSEN



About Nansen

Nansen's blockchain financial news service boosted customer engagement through real-time empowerment and an AI predictive strategy with Google Cloud solutions.

Named after Fridtjof Nansen, a Norwegian explorer, scientist, diplomat, and humanitarian, Nansen is a blockchain financial analytics platform. Their mission is to enable crypto investors to make informed decisions based on the highest quality data intelligence. The Singapore-based startup deploys AI algorithms to analyze petabytes of crypto trading data per day, allowing customers to discover opportunities, perform due diligence, and defend portfolios with real-time dashboards and alerts.

Google Cloud results

- Drives 20x increase in revenue over one year through BigQuery-powered data intelligence
- Enables real-time crypto data feed by crunching up to 1 petabyte of blockchain data per day
- Frees up two developer workloads for creative tasks through Cloud Composer workflow orchestration
- Inspires AI predictive modeling and recommendations strategy with BigQuery ML and Cloud Inference API

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Industries: Financial Services & Insurance

Location: Singapore

(<https://cloud.google.com/composer/>), [Cloud Run](https://cloud.google.com/run/) (<https://cloud.google.com/run/>)

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>), [Cloud Pub/Sub](https://cloud.google.com/pubsub/)

Cloud Composer

(<https://cloud.google.com/composer/>)

Cloud Run (<https://cloud.google.com/run/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

[Cloud Pub/Sub](https://cloud.google.com/pubsub/) (<https://cloud.google.com/pubsub/>)

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Crypto trading has seen dizzying growth in recent years, as investors flock to a new paradigm of decentralized finance (DeFi) enabled by blockchain technology. The value of DeFi recently surpassed \$100 billion

(<https://www.bloombergquint.com/onweb/defi-tops-100-billion-for-first-time-as-cryptocurrencies-surge>)

, up from \$4 billion

(<https://www.cnn.com/2021/09/17/how-decentralized-finance-works-and-why-its-taking-on-wall-street.html>)

in just four years.

With new opportunities come new complexities and risks such as market swings. Crypto assets are generated at a frenetic pace, so distinguishing quality from hype becomes challenging. Investors started searching for quality data intelligence for the DeFi universe, normally afforded by financial data services to big investment banks.

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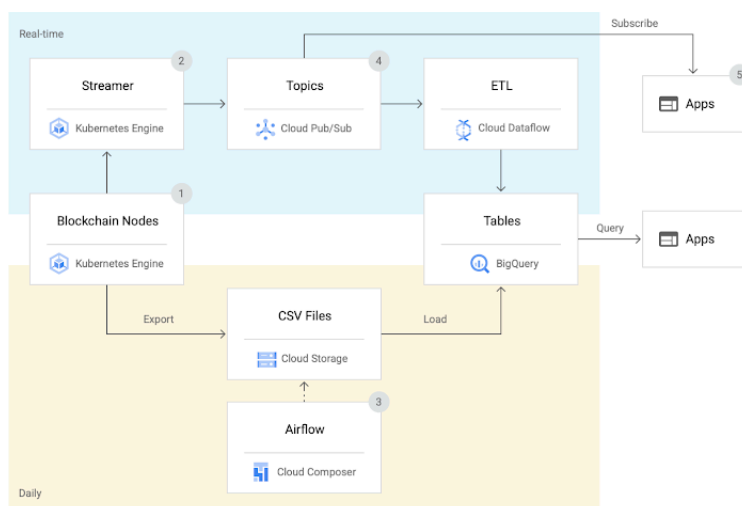
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—Alex Svanevik, co-
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Officer, Nansen

This formed the genesis for Nansen
(<https://www.nansen.ai/>) and its mission to surface
signals in blockchain data to the crypto community,
enabling traders to make informed bets from data
that truly matters. Founded by Alex Svanevik, Lars
Bakke Krogvig, and Evgeny Medvedev, Nansen
allows customers to discover opportunities such as
new-to-market tokens and analyze risk through real-
time notifications of significant events and trends in
the crypto arena.

To succeed, Nansen needed the most powerful data warehousing solution and scalable cloud infrastructure available to process the petabytes of trading data hitting its systems every day. It also required sophisticated data filters to send real-time smart alerts and personalized feeds so traders can stay ahead of the game. In turning to Google Cloud (<https://cloud.google.com/>) for auto-scaling blockchain infrastructure and managed data pipelines, Nansen could focus on building a product loved by customers at the vanguard of blockchain finance.

"For decentralized finance to become reality, it's important that the pioneers who enter the space succeed. Our mission is to support these pioneers and help them be winners," says Alex Svanevik, co-founder and Chief Executive Officer of Nansen. "With powerful solutions such as BigQuery (/bigquery), Cloud Run (/run), and Cloud Composer (/composer), we provide real-time intelligence to level the crypto playing field."



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—Evgeny Medvedev, co-founder & Chief Solutions Architect, Nansen

Uncovering powerful crypto currents for trading opportunity with BigQuery data analytics

Nansen feeds customers a real-time stream of investment signals that enables them to make quality decisions. One key way is discovering the identity of players, such as funds or exchanges that are moving crypto markets. Svanevik says this shows customers "where the smart money's going, what the best investors are buying."

Nansen puts a face to the trades through AI algorithms that make inferences on the identity of crypto movers and shakers, extracting easy-to-understand labels from the opaque wallets behind transactions. The system requires crunching staggering amounts of trading data openly available on blockchain, thus enabling the AI sleuthing that yields intelligence into the forces moving crypto trading.

"One critical reason why we chose Google Cloud over other providers is it focuses far more on data services compared to rivals," says Evgeny Medvedev, co-founder and Chief Solutions Architect at Nansen. "BigQuery simply doesn't have any competition in terms of data analytics features and autoscaling power. That, combined with the granular compute power of [Google Kubernetes Engine](#) (/kubernetes-engine) and Cloud Run, made our provider choice easy."

BigQuery enables Nansen to store 60 terabytes of crypto trading data and process up to one petabyte of data per day. Massive data power drives crypto intelligence that has contributed to a nearly 20-fold increase in revenue for the startup in 2021, along with an estimated 20x rise in its subscription rate.

As additional public blockchains come to market, Nansen can quickly iterate to add new chains and customer feeds leveraging its battle-tested blockchain node and data infrastructure on Google Cloud.

"I immediately fell in love with the intuitive user experience of Google Cloud. It was such a huge improvement over other providers that we were able to save two developer workloads."

—Evgeny Medvedev, co-founder & Chief Solutions Architect, Nansen

Curated data feeds for the crypto community enabled by custom

queries and data filtering

A prominent characteristic of DeFi is the variety and proliferation of crypto assets. This creates niche corners in the market inhabited by highly motivated players. For Nansen, that in turn creates a need to build highly personalized news feeds, tailored to the interests of each customer.

Even as Nansen requires auto-scaling compute and data analytics power to cover the entire global blockchain, it must also curate the 1% of relevant material that each individual subscriber wants from the data "firehose" published every day.

Google Cloud plays a key role in overcoming the challenge, shares Medvedev. The custom queries function on BigQuery enables Nansen to filter out irrelevant data in news feeds. Meanwhile, Pub/Sub (/pubsub) filters are the backend enabler of highly specialized and modular data streams that can be fed into BigQuery for instant machine learning insight.

Pinpointed data analytics allows Nansen to send smart alerts based on defined criteria, empowering customers with relevant intelligence. The results have been significant. Trial users activating at least one smart alert, says Svanevik, have a 400% higher chance of converting to full-fledged subscribers.

"Data filtering on BigQuery and Pub/Sub has been instrumental to us being able to build curated smart alerts for customers," says Svanevik. "That enabled

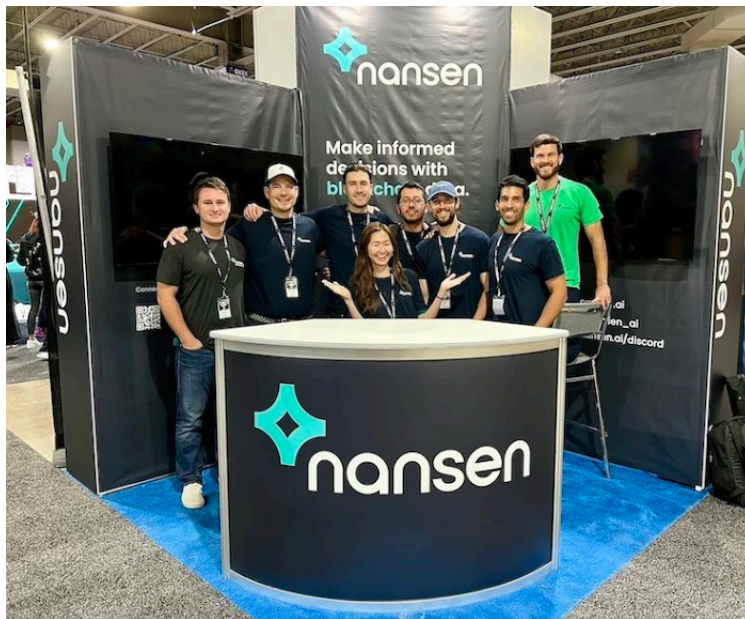
us to win a fourfold increase in our subscription conversion rate."

Cloud Composer, meanwhile, orchestrates workflows for all these complex data and computing processes, enabling engineers to focus on creative problems. Medvedev estimates Cloud Composer absorbs the workload of up to two engineers in Nansen's development team of 25.

Another contributor to the company's operational efficiencies is Google Cloud's "superior user experience." Having spent more than a decade working with different cloud providers, Medvedev says he felt an immediate impact when Nansen tapped Google Cloud, simply through better UX.

"I immediately fell in love with the intuitive user experience of Google Cloud. It was such a huge improvement over other providers that we were able to save two developer workloads," says Medvedev.

"The look and feel of the Google Cloud console is so similar to other Google services, such as Gmail (<https://workspace.google.com/products/gmail/>) and Google Maps (<https://www.google.com/maps>), making it natural and enjoyable to use."



Charting new AI-driven crypto paths with Google Cloud machine learning solutions

Nansen is now turning to the next stage in its evolution, developing AI-driven crypto predictive models. It also plans to create a recommendation system for crypto tokens based on similar purchase patterns by different users, a technique called collaborative filtering.

Medvedev says [BigQuery ML \(/bigquery-ml/docs\)](https://cloud.google.com/bigquery/ml/docs/), [Cloud Inference API \(/inference\)](https://cloud.google.com/ai/inference/), and [TensorFlow \(https://www.tensorflow.org/\)](https://www.tensorflow.org/) will play a strong role in developing these solutions that will add new dimensions to Nansen's journey of empowerment.

"We're only at the beginning of our adventure in achieving the dream of decentralized finance," concludes Svanevik. "With Google Cloud AI tools, we see a limitless future creating predictive modeling and recommendation systems to maximize our value to the global crypto community."

